

ECHOLIGHT

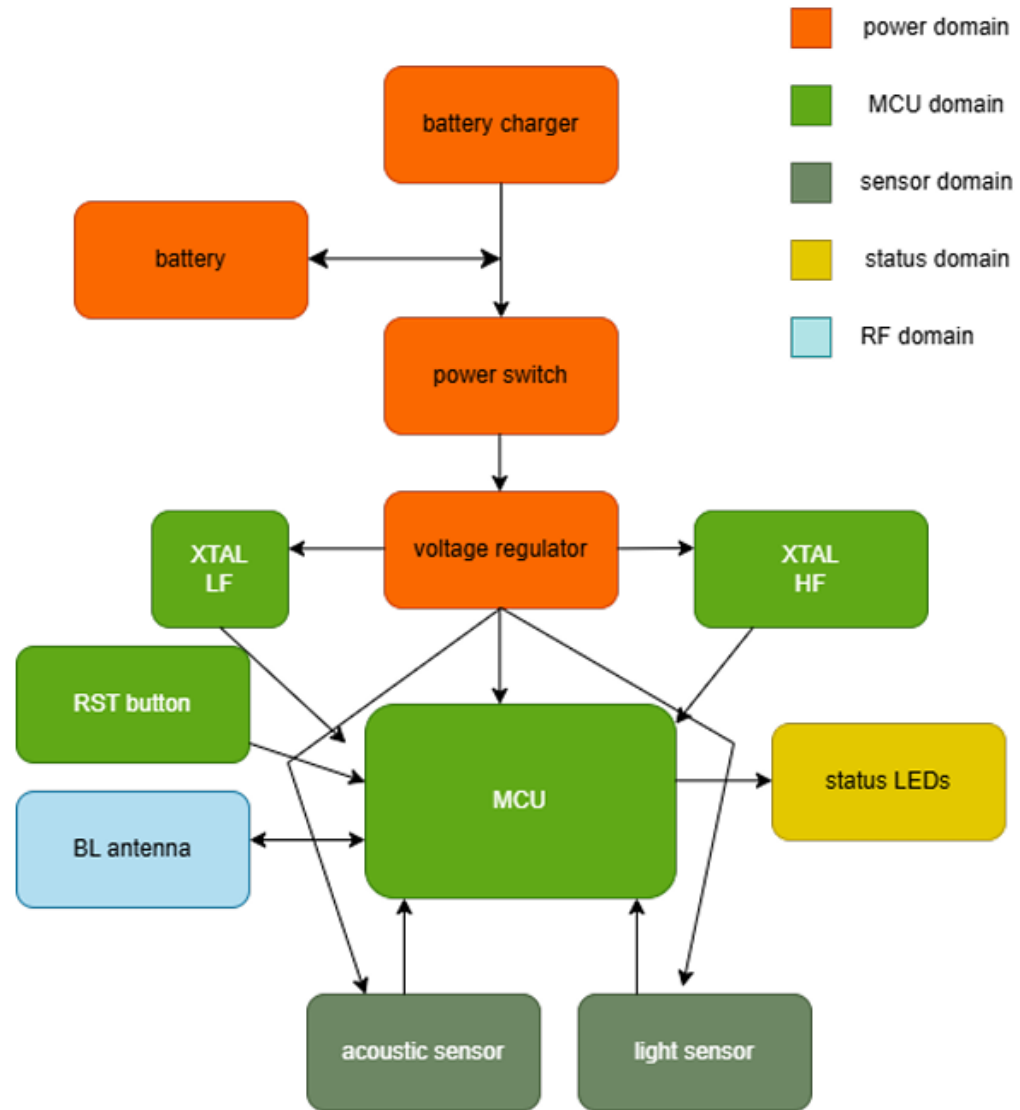
a dual sensor logger

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Project specifications

- Implementation of a dual logger that visualizes the electromagnetic field in the room and sends to the owner the state of sound and light
- Simultaneous acquisition and analysis of acoustic and lightning trend visualization
- Time alignment
- Stable multi-channel signal processing
- PCB physical and electrical isolation between channels to reduce interference: 4-layer PCB
- Reliable
- Usage: indoor applications (school, industrial, offices, museums) even in low light conditions

BLOCK DIAGRAM

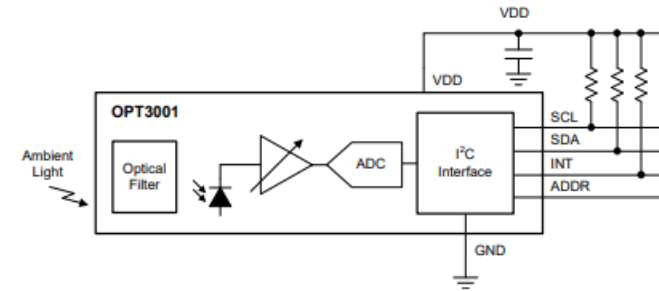


OPTICAL SENSOR: option 1

Texas instrument OPT3001 Ambient Light Sensor

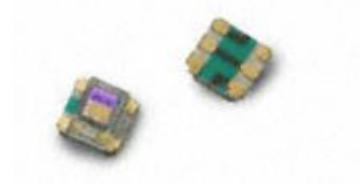


- Has a digital I2C interface
- Spectral response very close to the human-eye sensibility
- Operating current of 1.8 μA
- Supply voltage range: 1.6 V to 3.6 V
- Measurements on 23 bits: from 0.1 lux to 83 klux
- 1.98 €



OPTICAL SENSOR: option 2

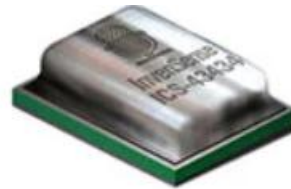
Broadcom:apds-9005



- current output: can be affected by the switching noise that is present on the board
- The analog measurement is strongly affected by the thermal noise that is present in the environment
- Spectral response close to the one of the human eye
- Supply current of 52 μA
- Supply voltage: 1.8-5.5 V
- only a resistor R_{load} of 5 kohm is needed for the conditioning
- 1.02 %
- 0.88 €

ACOUSTIC SENSOR: option 1

Invensense ICS-43434



- Microphone sends data through I2S interface (DMA access)
- I2S collects data while cpu is sleeping (wakes up more or less every 100 ms to process the data)
- Better quality because data is digitalized before entering the MCU, immune to Bluetooth noise
- SEPARATE FROM BLUETOOTH ANTENNA (TO AVOID EMI)
- SNR up to 64 dBA
- Frequency range: 60 Hz to 20 kHz
- 24-bit measurements
- 2,82 €

ACOUSTIC SENSOR: option 2

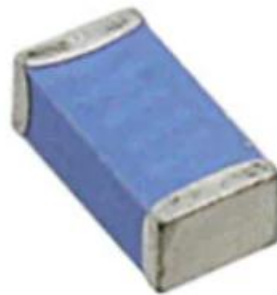
Same Sky: CMM-3125AT-42316-TR



- Analog microphone: can be subjected to the noise that is present in the environment
- Measured sound range: from 35 dB to 104 dB
- Frequency range: from 100 Hz to 10 kHz
- SNR: up to 65 dB
- Supply current: 250 μ A
- 0.84 €

ANTENNA: option 1

walsin technology corporation:RFANT3216120A5T

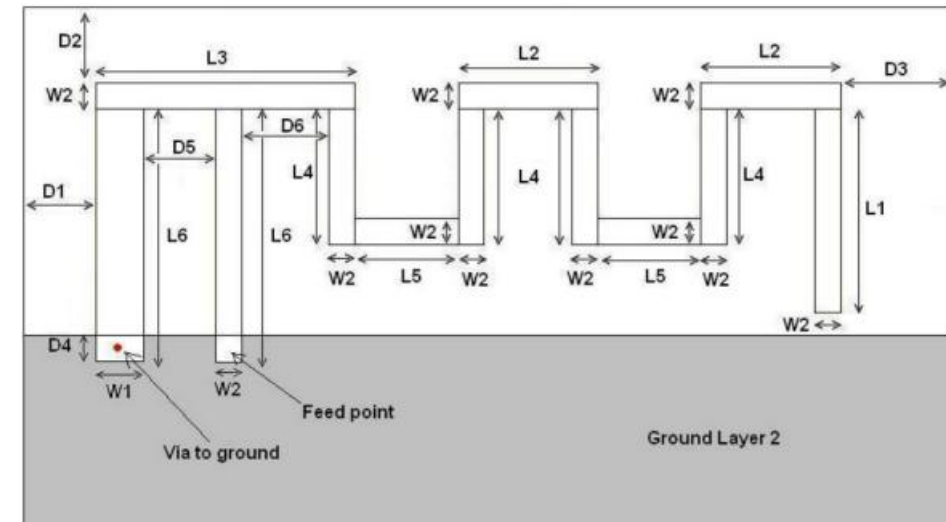


- Chip antenna: very small PCB area is occupied
- gain: 2 dBi
- Central frequency: 2.4 GHz, perfect for BL applications
- 0.3 €

ANTENNA: option 2

Patch antenna

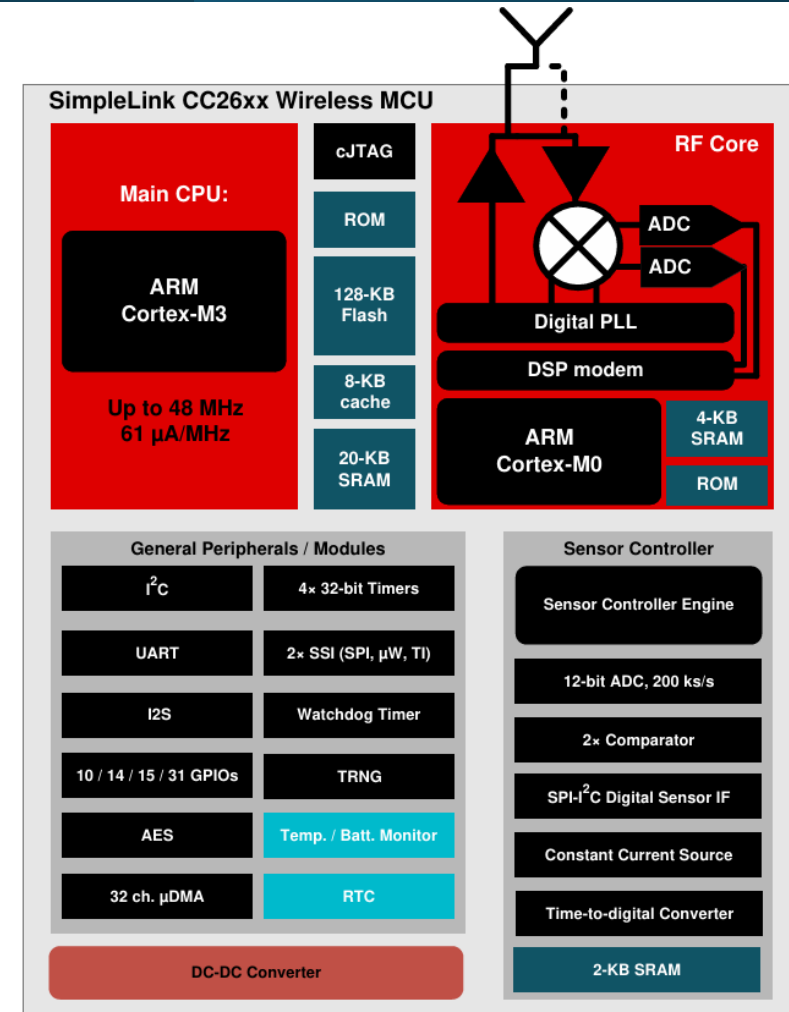
TOO MUCH AREA CONSUMING !!!!!



MICROCONTROLLER

Texas Instruments: CC2640R2FRHB

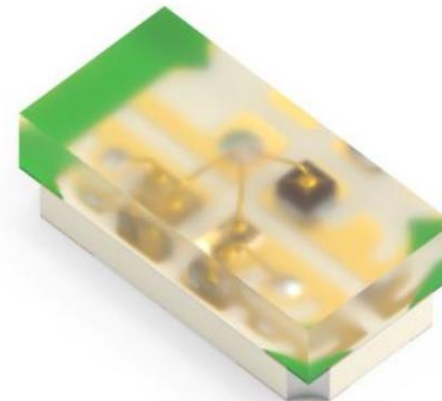
- 32 pins
- 4.00 mm x 4.00 mm
- Up to 48-MHz Clock Speed
- I2C, I2S
- Ultralow-Power Sensor Controller that can run autonomous from the rest of the system
- Sensor controller : capable of 1-Hz ADC sampling at 1 μ A system current
- 20KB of Ultralow-Leakage SRAM
- 2.4-GHz RF transceiver compatible with Bluetooth Low Energy 5.1
- 4.71 €



LED

SunLed: XZCMECBDDGK53W

- 1.6 x 0.8 mm
- RGB led
- Current: 2mA Red, 2mA Green, 2mA Blue
- Vf: 1.8V Red, 2.65V Green, 2.65V Blue
- 0.62 €



BATTERY

USE-702528-500PCBJST: 500 mAh



- LI-PO rechargeable battery
- Capacity: 500 mAh
- $V_{out} = 3.7\text{ V}$
- 5.49 €

LDO voltage regulator

Texas Instruments: TLV75801PDBVR



- adjustable LDO voltage regulator
- input voltage : from 1.5 V to 6 V
- I_{out_max} =500 mA
- quiescent current= 35 μ A
- For the regulation, only two resistors are required
- Output voltage: from 0.55 V to 5.5 V
- 0.23 €

Battery charger

ANALOG DEVICES:MAX1555



- DC charging current: 340 mA
- DC voltage range: 3.7-7 V
- DC to battery range: 0.1-6 V
- 2.50 €

POWER CONSUMPTION: contextualization

Microphone

Center-Aligned Snapshot: sample audio in the middle range of light conversion time. It produces 1 synchronized data point (Audio RMS + Light) per second → lower power + lower memory usage (100 ms window **centered within** the light integration period ~ 441 ms)

Light Sensor

Worst-case conversion time: **883 ms** (dark conditions)

MCU Processing data of audio samples

100 ms @ 51.6 kHz (mic max freq in high power mode) → **5160 samples** (16 bit / sample → 10.32 kB)

IIR + RMS: **10.75 ms** compute time assuming 100 clock cycles per sample

BLE Transmission

Payload: **2 Bytes (Audio RMS) + 2 Bytes (Light Data)**

Data + Overhead ~ 20 bytes. Active time @ 1 Mbps ~ 2-3 ms (being conservative)

Sensor Controller(SC)

Manages Light Sensor operations

Wakes twice (start/end)

Power estimation based on average consumption (1.0 μA in low power mode from datasheet)

LED Indicator : Resistors: **1 k Ω (Red), 100 Ω (Blue/Green)**

Green: **50% duty cycle** for 2 mins – then standby mode with **20% duty cycle** (other activities are negligible)

POWER CONSUMPTION: Timeline

Time (T)	Activity	Component Status	MCU State
0 ms	Start Cycle	Trigger Light Sensor (I2C Start)	Active (negligible)
0.1 ms	LED Check	Green LED ON	Sleeping
1 – 390 ms	Deep Sleep	Light: Integrating / Mic: OFF	Standby (SC active)
200 ms	LED Off	Green LED OFF	Standby
371 ms	Wake up	Wake Up Microphone (Stabilize)	Active
391 – 491 ms	Audio Snapshot	Mic ON :Recording / Light: Integrating	Sleep (DMA + I2S active)
491 ms	Stop Audio	Mic OFF / MCU Processing (RMS/Filter)	Active
501 – 880 ms	Wait Phase	Light: Integrating / Mic: OFF	Standby
883 ms	Light Ready	Light Sensor Interrupt (Read Data)	Active (SC standby)
885 ms	Transmit	BLE Radio TX (Send Data)	Active (RF)
890 – 1000 ms	Cycle Sleep	All Sensors OFF / Waiting for next sec	Standby

POWER CONSUMPTION: Timeline

$$I_{LightSensor} = \frac{T_{active} \cdot I_{active} + T_{sleep} \cdot I_{sleep}}{T_{total}} = \frac{883 \text{ ms} \cdot 3.7 \mu A + 117 \text{ ms} \cdot 0.4 \mu A}{1 \text{ s}} = \mathbf{3.3 \mu A}$$

$$I_{Microphone} = \frac{T_{active} \cdot I_{active} + T_{sleep} \cdot I_{sleep}}{T_{total}} = \frac{100 \text{ ms} \cdot 550 \mu A + 900 \text{ ms} \cdot 20 \mu A}{1 \text{ s}} = \mathbf{73.0 \mu A}$$

$$I_{MCU+BLE} = \frac{T_{proc} \cdot I_{proc} + T_{TX} \cdot I_{TX} + T_{stby} \cdot I_{stby}}{T_{total}} = \frac{10.75 \text{ ms} \cdot 2.93 \text{ mA} + 3 \text{ ms} \cdot 9.1 \text{ mA} + 986.25 \text{ ms} \cdot 1 \mu A}{1 \text{ s}} = \mathbf{59.78 \mu A}$$

$$I_{SC+DMA+I2S} = \frac{N_{wake \text{ up}} \cdot I_{SC} + T_{DMA} \cdot I_{DMA} + T_{I2S} \cdot I_{I2S}}{T_{total}} = \frac{2 \cdot 1.1 \mu A + 100 \text{ ms} \cdot 130 \mu A + 100 \text{ ms} \cdot 36 \mu A}{1 \text{ s}} = \mathbf{18.8 \mu A}$$

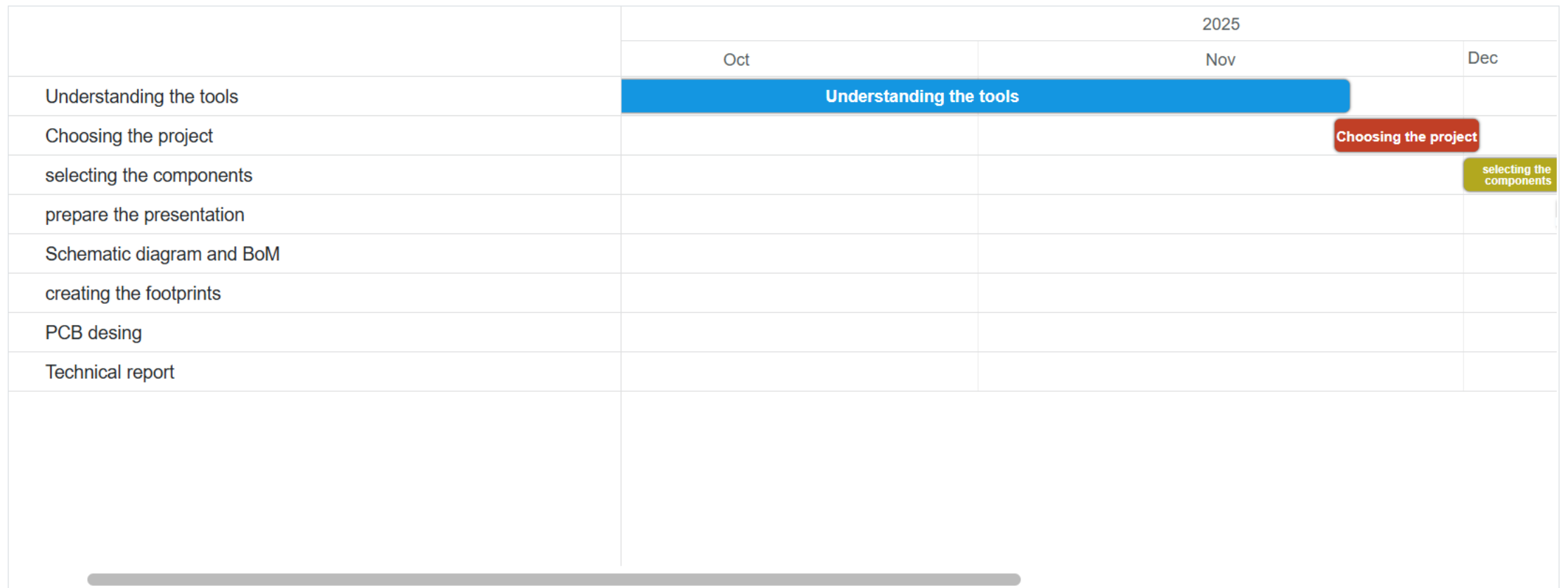
$$I_{LED} = \frac{T_{pulse} \cdot I_{ON}}{T_{period}} = \frac{200 \text{ ms} \cdot 20 \text{ mA}}{1 \text{ s}} = \mathbf{4.0 \text{ mA}}$$

$$\begin{aligned} BatteryLife &= \frac{BatteryCapacity}{I_{Mic} + I_{Light} + I_{MCU} + I_{LED} + I_{SC+DMA+I2S}} = \frac{500 \text{ mAh}}{73.0 \mu A + 3.3 \mu A + 59.78 \mu A + 4.0 \text{ mA} + 18.8 \mu A} = \\ &= \frac{500 \text{ mAh}}{4.1582 \text{ mA}} \approx \mathbf{120.35 \text{ h} \approx 5 \text{ days}} \end{aligned}$$

TOTAL PRICE

• Optical sensor OPT3001:	1.98 €
• Acoustic sensor ICS-43434:	2.82 €
• Antenna RFANT3216120A5T:	0.3 €
• MCU CC2640R2FRHB:	4.71 €
• LED XZCMECBDDGK53W :	0.62 €
• Passive components + switch:	3.51 €
• Battery USE-702528-500PCBJST 500 mAh:	5.49 €
• LDO TLV75801PDBVR:	0.23 €
• Battery charger MAX1555:	2.50 €
• PCB (4 layers)	5.19 €
• TOTAL:	26.99 €

GANTT DIAGRAM



MARKET ANALYSIS AND EXISTING SOLUTION

No light + noise dual logger was found, but market analysis was done by checking existing solutions based on separate loggers of light and noise.

- Light logger: HOBO Temp/RH/Light/Ext-Analog Data Logger
- Bluetooth-enabled logger
- €212.51

- Noise logger: Data Logging Sound Level Meter PCE-323
- Bluetooth-enabled logger
- € 305.71